

NIXON NG

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PROFILE

Data Engineer with over 3 years of consultant experience transforming commercial data ecosystems. Slashed migration timelines by 67% while ensuring 99.99% data accuracy for FTSE 100 client. Architected and implemented scalable ETL/ELT pipelines for processing 20GB+ daily across AWS and Azure. Spearheaded data teams on solution/model implementation for GenAI-powered chatbot and data products. Recently completed MSc Data Science to deliver next-generation solutions.

Personal/Work-related projects available on <https://github.com/Nixonngzh/Portfolio-Guide>

PROFESSIONAL EXPERIENCE

Senior Data Engineer, PwC | Kuala Lumpur, MY **Feb 2026- Current**

- Transformed Executive-level reporting towards AI Adoption by leveraging **Microsoft Azure/Power Platform** (i.e. integrating Document Intelligence Agents, Spec-Driven Analytics Webapp Development & Performance Advisor AI Chatbot)
- Assisted QA/Design of **PowerBI** dashboards for leadership users to monitor live/historical company offering performances
- Consulting company data migration (**SQL Server** to **Databricks Unity Catalog**) for Big Data Analytics and AI/ML workloads

Data Engineering Intern, Valor Real Estate Partners | London, UK **Sept 2025- Nov 2025**

- Deployed **Python** web-scrappers to capture 90% of European real estate market using API integrations (REST/JSON)
- Aided the Research & Strategy team to ingest financial/geospatial data into EU portfolio tableau dashboards using **Python**, **Snowflake** & **Dbt** for Box folder scraping, data warehousing and orchestration needs
- Conducted last mile modelling for Milan market using QGIS, Smappen and Maps API to inform investment decisions

Data Engineering Consultant, Kubrick Group | London, UK **Jan 2023- Sept 2024**

Overview

- Designed and improved **ETL pipelines** using automated **Airflow** jobs for Airlines client, resulting in 300% increase in data loading/checks efficiency for British Airways product suite. (i.e. Avios Hotels, Wine Flyer, Avis Carhire)
- Orchestrated Airlines customer ingestion from CRM sources using **Github Actions** and AWS-managed **Python/SQL** scripts
- Automated Insurance data ingestion, by developing batch **Snowpark ELT** jobs to integrate with client's AWS infrastructure

Data Engineer, IAG Loyalty (Client)

- Performed data analysis and developed Avios Customer Loyalty metrics dashboards for C-Suite decision-making, by ensuring data error rate below 0.01% through automated validation using SQL, **Tableau** and **Matillion ETL**
- Productionized **Airflow** DAGs on Migration data refreshes of 20GB/day from legacy **Teradata** system to **AWS S3/Snowflake** by using **Matillion ETL**, **Python** scripts and **AWS Services** (EC2, Glue & Lambda triggers)
- Developed high-level **DataVault** models, **Mapping** documents and **Validation** framework for 4 European Airlines back-end (British Airways, Iberia, Vueling, Aer Lingus) via **Atlassian** workflow and cross-regional collaboration

Data Project Lead (Kubrick Internal)

- Led data team of 5 to develop AI chatbot using **ChatGPT**, **Langchain**, & **Neo4j**, democratizing company knowledge stream
- Automated **Dbt** pipelines for data transformation, cleaning, lineage and quality checks on **Snowflake**
- Generated synthetic data using **Python** to represent **Salesforce** CRM feed into **Snowflake** through **Azure Cloud**

Research Analyst, University of Bath | Bath, UK **Mar 2021- Aug 2021**

- Conducted geospatial and time-series forecasting to analyze **COVID-19 transmission** across UK cities
- Created interactive dashboards in **Tableau** to visualize insights for academic research

EDUCATION

City University of London, MSc in Data Science **Sept 2024- Sept 2025**

- Awarded Distinction** - Big Data, Deep Learning (AI), Natural Language Processing, Computer Vision, Visual Analytics
- Master's Thesis:** Achieved 75% - Implemented Domain Adaptation on YOLO models in Multi-Object Tracking System for Broadcast Badminton Footage, demonstrating improved model accuracy/robustness on visually dynamic footages
- Technologies:** Python, MatLab, Apache Hadoop & Spark, Google Cloud Platform (GCP), Tableau

University of Bath, MEng in Chemical Engineering **Sept 2017- Jul 2022**

- Awarded Upper Second Class** - Engineering Product Design & Real-Time Python Simulation, Process Modelling and Control

A-levels: Chemistry A*, Physics A, Mathematics A, Economics B

Jul 2015- Jul 2017

SKILLS AND CERTIFICATIONS

Skills

- Programming Languages:** Python (Pandas, PySpark, Tensorflow), SQL, Cypher, JavaScript fundamentals
- Data Engineering:** ETL/ELT framework, Dimensional/Data Vault Modeling, Matillion, Dbt, Apache Spark & Kafka
- Big Data & Analytics:** Apache Hadoop, Tableau, PowerBI, Neo4j, BigQuery, Looker Studio, Grafana, QGIS, Excel
- Databases, Warehouses & Cloud:** PostgreSQL, Salesforce, Snowflake, Databricks, AWS, Microsoft Azure, GCP
- DevOps:** Github Actions, Azure DevOps, Bitbucket, Docker, Kubernetes, Apache Airflow, Jira, Confluence

Certifications

- Databricks Certified Data Engineer Professional (In Progress), Snowflake Hands-on Essentials, Neo4J Certified Professional